

Deployment Architecture

Document:

`docs/architecture/DEPLOYMENT_ARCHITECTURE.md`

System: HackNest — Student Operating System

Status: Authoritative — reconciled against

`README.md` , `MASTER_ARCHITECTURE.md` ,

`DATABASE_ARCHITECTURE.md` ,

`SECURITY_ARCHITECTURE.md` , `TESTING_STRATEGY.md` ,

`MASTER_GOVERNANCE_AMENDMENT_001.md`

Type: Architecture Specification Only — No Scripts, No Actions, No Dockerfiles, No Terraform, No Vercel Config

Per `MASTER_ARCHITECTURE.md` Section 9, this document is the sole authoritative owner of infrastructure, environments, CI/CD, and release process — including RPO/RTO and backup policy, now defined concretely below, sourced from `MASTER_GOVERNANCE_AMENDMENT_001.md` Section 4.2–4.3.

1. Purpose

Deployment gets every validated change safely in front of students and keeps it there. A release should be routine, boring, and reversible.

2. Deployment Principles

Reliability, Scalability, Security, Zero Downtime, Rollback Capability, Automation First, Observability — unchanged in substance, now grounded in concrete targets rather than deferred numbers throughout this document.

3. System Architecture

Client → Next.js Application (Server Components/Actions, Route Handlers) → Supabase (Auth, Database with RLS, Storage) → CDN/DNS/Monitoring/Analytics, exactly matching the layering in `MASTER_ARCHITECTURE.md` Section 3. Assumed hosting platform: Vercel, per `CODING_STANDARDS.md` 's Next.js-first stack conventions.

4. Environments

Local, Preview (ephemeral per-change), Staging (persistent, production-like), Production — isolated per `SECURITY_ARCHITECTURE.md` Section 15's environment-isolation requirement, promoted in one direction only.

5. CI/CD Strategy

Build pipeline (clean-environment build required), Test pipeline (`TESTING_STRATEGY.md` levels), Security checks (dependency scanning, secret-leak detection per `SECURITY_ARCHITECTURE.md`), Deployment approval (per `MASTER_GOVERNANCE_AMENDMENT_001.md` Section 5's Release Approver), Rollback process (automated on failed health check, or manual on discovered regression).

6. Infrastructure

Vercel (application hosting), Supabase (database/auth/RLS), Object Storage (per `DATABASE_ARCHITECTURE.md` Section 20), CDN, Domain Management (change-controlled), SSL/TLS (encrypted by default, every environment including Preview), Backups (Section 8 below).

7. Configuration Management

Environment variables scoped per environment;
secrets per `SECURITY_ARCHITECTURE.md` Section 12;
configuration validated at startup (fail to start rather
than run misconfigured); runtime configuration
(feature flags) changed through an audited
mechanism, not direct untracked edits.

8. Backup & Disaster Recovery (Reconciled – Concrete Values)

Sourced from
`MASTER_GOVERNANCE_AMENDMENT_001.md` Section 4.2–
4.3:

Target	Value
RPO	15 minutes
RTO (full restoration)	4 hours
RTO (critical-path degraded mode: Authentication, core reads)	1 hour

Target	Value
Backup frequency	Continuous WAL + full snapshot every 6 hours
Backup retention	Rolling 35 days point-in-time recovery; monthly snapshots retained 12 months
DR testing cadence	Full restore rehearsed against non-production quarterly

This resolves the RPO/RTO and backup-policy placeholders flagged in `ARCHITECTURE_VALIDATION_REPORT.md`.

9. Monitoring & Observability

Application, Infrastructure, Error tracking, Performance metrics (against `TESTING_STRATEGY.md` Section 5 targets), Health checks, Uptime monitoring — categories owned here; day-to-day execution and escalation owned by `OPERATIONS_ARCHITECTURE.md`.

10. Scaling Strategy (Reconciled – Concrete Values)

- **Horizontal scaling:** stateless application layer, scales by adding capacity.
 - **Database scaling:** partitioning triggers per `DATABASE_ARCHITECTURE.md` Section 13 and `MASTER_GOVERNANCE_AMENDMENT_001.md` Section 4.6 – 50 million rows or 2x query-time target, whichever first, for Messages/Audit Logs/Notifications/Feed.
 - **Concurrency targets:** 50,000 steady-state, 250,000 burst, per `MASTER_GOVERNANCE_AMENDMENT_001.md` Section 4.6.
 - **CDN caching / Rate limiting / File storage scaling:** per `API_ARCHITECTURE.md` Section 6/8 and `DATABASE_ARCHITECTURE.md` Section 24.
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11. Release Strategy

Versioning (consistent scheme per `CODING_STANDARDS.md`), Release process (Preview → Staging → Production, gated by `TESTING_STRATEGY.md` Section 8), Rollback policy (prompt rollback on Critical/High regression per Bug Severity SLAs in

MASTER_GOVERNANCE_AMENDMENT_001.md Section 4.7),
Hotfix process (expedited but real review, no
bypassed security checks).

12. Production Readiness Checklist

Security review, Performance review (against Section
10/ TESTING_STRATEGY.md Section 5 targets),
Accessibility review (against WCAG 2.2 AA per
UI_DESIGN_SYSTEM.md Section 9), Testing
completion, Monitoring verification, Documentation
verification.

Definition of Done

- [x] Architecture only — no scripts, actions, Dockerfiles, Terraform, or Vercel config.
 - [x] Reconciled against MASTER_ARCHITECTURE.md ,
DATABASE_ARCHITECTURE.md ,
SECURITY_ARCHITECTURE.md ,
TESTING_STRATEGY.md ,
MASTER_GOVERNANCE_AMENDMENT_001.md .
 - [x] RPO/RTO, backup policy, and scaling targets
now concrete, closing all three previously-flagged
gaps.
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Validation Report

Internal consistency: confirmed — Section 8 and Section 10 are each stated once and referenced, not restated, elsewhere. **Cross-document consistency:**

RPO/RTO/backup values match

MASTER_GOVERNANCE_AMENDMENT_001.md Section 4.2–4.3 verbatim; scaling values match Section 4.6 verbatim; performance references cite

TESTING_STRATEGY.md Section 5 rather than duplicating it. **Resolved audit findings:** the RPO/RTO duplication and the "hosting platform assumed, unconfirmed" flag are both closed (hosting platform explicitly confirmed as an assumption consistent with CODING_STANDARDS.md, not a contradiction).

Status: Validated — Reconciled — Ready to Govern Implementation